CS 2300 HW11

1) Let us assume 3 cases for this question: Leftover banana, leftover orange and leftover apple.

Case 1: Leftover banana

Since we considered that a banana was left, we are left with 4 bananas right now and the number of oranges and apples remains the same, so the number of ways in which we can distribute the remaining fruits can be calculated in the following way:

11!/4!*3!*4! = 11550

Therefore,we have 11550 ways in which we can distribute the remaining fruits between 11 friends.

Case 2: Leftover Orange

Since we considered that a orange was left, we are left with 2 oranges right now and the number of bananas and apples remains the same, so the number of ways in which we can distribute the remaining fruits can be calculated in the following way:

11!/5!*2!*4! = 6930

Therefore,we have 6930 ways in which we can distribute the remaining fruits between 11 friends.

Case 3: Leftover Apple

Since we considered that a apple was left, we are left with 2 apples right now and the number of oranges and bananas remains the same, so the number of ways in which we can distribute the remaining fruits can be calculated in the following way:

11!/5!*3!*3! = 9240

Therefore,we have 9240 ways in which we can distribute the remaining fruits between 11 friends.

2) We can denote the number of ways in which 24 distinct items can be divided into groups(ordered) each having a size of 3 by:

24!/3!*3!*3!*3!*3!*3!*3!*3! = $24!/(3!)^8$

Over here, we have 8 factors of 3!, so that's why the denominator can be denoted as $(3!)^8$.

a) Since each box is distinguishable over here, we can simply assign 3 books to box 1, other 3 books to box 2 and the next 3 books to box 3 and so on:

$$(24!/(3! * 18!)) * 21!/(3! * 15!)) *.... = 24!/(3!)^8$$

b) Over here, since the boxes are indistinguishable right now, we can divide it further by 8! So that we can find the number of ways to pack 24 different books into 8 boxes where each box can contain 3 books:

$$(1/8!) * 24!/(3!)^8 = 24!/(8! * 3!^8)$$

c) Let us first try and partition all 24 books into 8 unlabeled groups of 3:

$$24!/((3!)^8 * 8!) \text{ (i)}$$

Next, let us choose which 4 of those 8 groups are supposed to be shipped, so we can do that by:

$$8!/(4! * 4!) \text{ (ii)}$$

Let us label each of the shipped boxes by an address. Since there are 3 possible addresses for each box, we can denote it by 3^4(iii).

Finally, in order to take out the number of ways in which 4 boxes are supposed to shipped while leaving the other 4 boxes can be calculated by multiplying (i),(ii) and (iii):

$$24!/(8! * (3!)^8) * 8!/(4! * 4!) * 3^4 = (3^4 * 24!)/(3!)^8 * (4!)^2)$$

3) Over here we can say that $x_1 + x_2 + x_3 + x_4 = 12$, where x_i is the number of cookies a person i gets.

a) No restriction

Since there is no restriction over here, we can assume that x_i could be 0 or even greater than that for any person i. Hence, we can count the number of ways in which we can distribute 12 cookies amongst 4 people can be calculated as follows:

(12+4-1)!/((4-1)! * (12+4-1-4+1)! = 15!/(3!*12!) = 455

Therefore, there are 455 ways in which 12 cookies can be distributed amongst 4 people with no restriction.

b) **Each kid gets at least 2 cookies**

Let's say that we have another variable y_i and that is equal to $x_i - 2$, then we can calculate the summation of y_i by:

$$\sum y_i = \sum x_i - (2*4)$$

Since $\sum x_i$ = 12, this can be written as:

$$\sum y_i = 12 - (2*4) = 4$$

Now, we can calculate the number of ways in which the cookies can be distributed for this condition by:

(4+4-1)! / (4-1)! (4+4-1-4+1)! = 7! / (3! * 4!) = 35

Therefore, there are 35 ways in which 12 cookies can be distributed amongst 4 people so that each person gets at least 2 cookies.

c) Let us assume that $x_1 = Sam,\ x_2 = John\ and\ x_3\ and\ x_4\ are\ others$. Hence, we can say that $y_1 = x_1 - 4$ (>=0), 0 <= x_2 <=4 and x_3 and x_4 are greater than or equal to 0, so the total becomes $y_1 + x_2 + x_3 + x_4 = 12 - 4 = 8$. Therefore, we sum over John's possible x_2 = j = 0,1,2,3,4.For each j, the remaining will be $y_1 + x_3 + x_4 = 8 - j$ which has:

((8-j)+3-1)!/(3-1)!*(8-j-3+1) = (10-j)!/2!*(6-j)! Solutions

Hence, we can calculate the total by doing its summation:

∑ (10-j)!/2!*(6-j)! = 10!/2!*(6!) + 9/2!*5! + 8!/2!*4! + 7!/2!*3! + 6!/2!*2! = 45 + 36+28+21+15 = 145 (We will take the summation from j=0 to j=4)

Therefore, there are 145 ways in which Sam gets at least 4 cookies and John gets at most 4 cookies from 12.

4a) As boxes are unlabeled and the books are indistinguishable, we can that a configuration could be written as:

6 = n1 + n2 +….+n5

Over here, n_i >=0 and the order of n_i doesn't matter. The number of partitions of 6 into at most 5 parts can be denoted as $p_5(6)$. We can say that p(6) = 11 for all partitions of 6, but, since exactly one of those uses 6 parts, that's why when we look into at most 5 parts there are 10 ways.

b) Now, over here the case is a bit different, the books are distinct but the boxes are unlabeled. We can say that the empty blocks are irrelevant once they are identical, so basically it's just the number of set partitions with 1<=k<=5 blocks and we can represent its sum as follows:

∑ S(6,k) = $B_6 - S(6,6)$ = 203 -1 = 202(Over here, we take the sum from k=1 to k=5)

Over here, we have:

S(6,k): It is the stirling number of the second kind
B_6 = the summation of S(6,k) from k = 0 to k = 6 = 203, which is the 6th bell number.
S(6,6) = 1, which is the partition into 6 singletons.

5a) Over here, we basically need to decide the number of books that go into each shelf, which can be calculated in the following way:

(8+5-1)!/(5-1)!(8+5-1-5+1)! = 12!/(4!*8!) = 495 ways

Therefore, there are 495 ways in which we can keep 8 books on a bookcase with 5 shelves, provided that the books are indistinguishable from each other.

b) Let us first try and line up 8 books in one of 8! Orders, after that we can choose 4 cut-points among the 12 gaps (which can be calculated by doing 8+5-1) to break it into 5 shell segments. From this, we get that:

8! * (12!/(4!*8!)) = 40320 * 495 = 19958400 ways

Therefore, there are 19958400 ways in which we can keep 8 books on a bookcase with 5 shelves, provided that all books are distinct.

6) Let us calculate the sum of all the degrees first.

5 + 4 + 3 +3 + 2+ 2 = 19

Since, we know that 19 is an odd number and by Handshake Lemma, the sum of all the degrees should be even, that is, twice the number of edges, but since it's odd, we can say that no such graph exists.

7a) No, the graph is not connected and we can prove this by providing an example as follows.

Counterexample

Let us consider a cycle on n-1 vertices, which has n-1 edges, plus one isolated vertex. So, we can say that the total number of edges is n-1, but the graph is not connected.

b) No, the graph is not acyclic and we can prove this by providing an example.

Counterexample

Let us say that for n = 4, we have a 3-cycle on {1,2,3} and an isolated vertex 4 has 3 edges (4-1) but it contains a cycle.

8) **Claim:** An undirected graph T is a tree, which means that it is connected and acyclic, if T is acyclic and adding any new edge to T creates a cycle.

Proof

"⇒"If T is a tree, it means that it has no cycles and is connected.So, for any two non adjacent vertices u and v connectivity gives a path P between them, which adds the edge uv that closes P into a cycle.

"⇐", Let us consider that G is acyclic and at maximal with respect to any extra edges that induces a cycle, G was connected, it picks vertices u and v in different components by adding the edge uv will not create any cycle contradiction. Hence we can say that G is connected and since it is acyclic, we can say that G is a tree.